

An Analysis of Hidden Markov model

Hidden Markov model

$$P(O | \lambda) = \sum_Q P(O | Q, \lambda) P(Q | \lambda)$$

Hidden Markov model -- Part 1

The choice of the urn for the n -th ball depends only upon a random number and the choice of the urn for the $(n - 1)$ -th ball. The choice of urn does not directly depend on the urns chosen before this single previous urn. Therefore, this is called a Markov process. It can be described by the upper part of Figure 1.

Hidden Markov model -- Part 2

Most likely explanation The task, unlike the previous two, asks about the joint probability of the entire sequence of hidden states that generated a particular sequence of observations (see illustration on the right). This task is generally applicable when HMM's are applied to different sorts of problems from those for which the tasks of filtering and smoothing are applicable. An example is part-of-speech tagging, where the hidden states represent the underlying parts of speech corresponding to an observed sequence of words. In this case, what is of interest is the entire sequence of parts of speech, rather than simply the part of speech for a single word, as filtering or smoothing would compute. This task requires finding a maximum over all possible state sequences, and can be solved efficiently by the Viterbi algorithm.

Hidden Markov model -- Part 3

In its discrete form, a hidden Markov process can be visualized as a generalization of the urn problem with replacement (where each item from the urn is returned to the original urn before the next step).

Hidden Markov model -- Part 4

Concepts Teif, V. B.; Rippe, K. (2010). "Statistical-mechanical lattice models for protein-DNA binding in chromatin". *J. Phys.: Condens. Matter.* 22 (41) 414105. arXiv:1004.5514. Bibcode:2010JPCM...22O4105T. doi:10.1088/0953-8984/22/41/414105. PMID 21386588. S2CID 103345. A Revealing Introduction to Hidden Markov Models by Mark Stamp, San Jose State University. Fitting HMM's with expectation-maximization – complete derivation A step-by-step tutorial on HMMs Archived 2017-08-13 at the Wayback Machine (University of Leeds) Hidden Markov Models (an exposition using basic mathematics) Hidden Markov Models (by Narada Warakagoda) Hidden Markov Models: Fundamentals and

Applications Part 1, Part 2 (by V. Petrushin) Lecture on a Spreadsheet by Jason Eisner, Video and interactive spreadsheet

Hidden Markov model -- Part 5

In this piece of code, `start_probability` represents Alice's belief about which state the HMM is in when Bob first calls her (all she knows is that it tends to be rainy on average). The particular probability distribution used here is not the equilibrium one, which is (given the transition probabilities) approximately {'Rainy': 0.57, 'Sunny': 0.43}. The `transition_probability` represents the change of the weather in the underlying Markov chain. In this example, there is only a 30% chance that tomorrow will be sunny if today is rainy. The `emission_probability` represents how likely Bob is to perform a certain activity on each day. If it is rainy, there is a 50% chance that he is cleaning his apartment; if it is sunny, there is a 60% chance that he is outside for a walk.

Hidden Markov model -- Part 6

Terminology The states of the process X_n (resp. X_t) are called hidden states, and $\{Y_n \in A \mid X_n = x_n\}$ (resp. $\{Y_t \in A \mid X_t = x_t\}$) is called emission probability or output probability.

Hidden Markov model -- Part 7

Filtering The task is to compute, given the model's parameters and a sequence of observations, the distribution over hidden states of the last latent variable at the end of the sequence, i.e. to compute $P(x_t \mid y(1), \dots, y(t))$. This task is used when the sequence of latent variables is thought of as the underlying states that a process moves through at a sequence of points in time, with corresponding observations at each point. Then, it is natural to ask about the state of the process at the end. This problem can be handled efficiently using the forward algorithm. An example is when the algorithm is applied to a Hidden Markov Network to determine $P(h_t \mid v_{1:t})$.

Hidden Markov model -- Part 8

A hidden Markov model (HMM) is a Markov model in which the observations are dependent on a latent (or hidden) Markov process (referred to as X). An HMM requires that there be an observable process Y whose

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outcomes depend on the outcomes of X in a known way. Since X cannot be observed directly, the goal is to learn about state of X by observing Y . By definition of being a Markov model, an HMM has an additional requirement that the outcome of Y at time $t = t_0$ must be "influenced" exclusively by the outcome of X at $t = t_0$ and that the outcomes of X and Y at $t < t_0$ must be conditionally independent of Y at $t = t_0$ given X at time $t = t_0$. Estimation of the parameters in an HMM can be performed using maximum likelihood estimation. For linear chain HMMs, the Baum–Welch algorithm can be used to estimate parameters. Hidden Markov models are known for their applications to thermodynamics, statistical mechanics, physics, chemistry, economics, finance, signal processing, information theory, pattern recognition—such as speech recognition, handwriting recognition, gesture recognition, part-of-speech tagging, musical score following, partial discharges and bioinformatics.

Hidden Markov model -- Part 9

Other extensions Yet another variant is the factorial hidden Markov model, which allows for a single observation to be conditioned on the corresponding hidden variables of a set of K independent Markov chains, rather than a single Markov chain. It is equivalent to a single HMM, with N^K states (assuming there are N states for each chain), and therefore, learning in such a model is difficult: for a sequence of length T , a straightforward Viterbi algorithm has complexity $O(N^{2K}T)$. To find an exact solution, a junction tree algorithm could be used, but it results in an $O(N^{K+1}KT)$ complexity. In practice, approximate techniques, such as variational approaches, could be used. All of the above models can be extended to allow for more distant dependencies among hidden states, e.g. allowing for a given state to be dependent on the previous two or three states rather than a single previous state; i.e. the transition probabilities are extended to encompass sets of three or four adjacent states (or in general K adjacent states). The disadvantage of such models is that dynamic-programming algorithms for training them have an $O(N^KT)$ running time, for K adjacent states and T total observations (i.e. a length- T Markov chain). This extension has been widely used in bioinformatics, in the modeling of DNA sequences. Another recent extension is the

triplet Markov model, in which an auxiliary underlying process is added to model some data specificities. Many variants of this model have been proposed. One should also mention the interesting link that has been established between the theory of evidence and the triplet Markov models and which allows to fuse data in Markovian context and to model nonstationary data. Alternative multi-stream data fusion strategies have also been proposed in recent literature, e.g., Finally, a different rationale towards addressing the problem of modeling nonstationary data by means of hidden Markov models was suggested in 2012. It consists in employing a small recurrent neural network (RNN), specifically a reservoir network, to capture the evolution of the temporal dynamics in the observed data. This information, encoded in the form of a high-dimensional vector, is used as a conditioning variable of the HMM state transition probabilities. Under such a setup, eventually is obtained a nonstationary HMM, the transition probabilities of which evolve over time in a manner that is inferred from the data, in contrast to some unrealistic ad-hoc model of temporal evolution. In 2023, two innovative algorithms were introduced for the Hidden Markov Model. These algorithms enable the computation of the posterior distribution of the HMM without the necessity of explicitly modeling the joint distribution, utilizing only the conditional distributions. Unlike traditional methods such as the Forward-Backward and Viterbi algorithms, which require knowledge of the joint law of the HMM and can be computationally intensive to learn, the Discriminative Forward-Backward and Discriminative Viterbi algorithms circumvent the need for the observation's law. This breakthrough allows the HMM to be applied as a discriminative model, offering a more efficient and versatile approach to leveraging Hidden Markov Models in various applications. The model suitable in the context of longitudinal data is named latent Markov model. The basic version of this model has been extended to include individual covariates, random effects and to model more complex data structures such as multilevel data. A complete overview of the latent Markov models, with special attention to the model assumptions and to their practical use is provided in